A LOCALIZED CLASS COMPUTED ON A CLOSED SUBSCHEME AGREES WITH THE CLASS OF A PERFECT OBJECT RESTRICTING TO THE SAME THING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FB8 (sources/stacks-chow/pinned-original-source.tex line 10975).

Allowed standard prerequisites (editorial).

- The classes P'_p and c'_p on a closed subscheme and their characterisation. Same chapter, Lemma labelled lemma-silly, cited at proof lines 10974-10980 and 11000-11002. Supplies the characterisation that the proof verifies for the class of E.
- The localized Chern class of a perfect object and its behaviour under shrinking the support. Same chapter, Definition labelled definition-localized-chern, invoked in the statement, and Lemma labelled lemma-loc-chern-shrink-Z, cited at proof lines 10994-10998. Writes P_p(X_2 -> X, E) through the class on the complement Z of U.
- Lifting the vanishing condition from X_1 to an open neighbourhood. More on Algebra, Lemma labelled more-algebra-lemma-lift-perfect-from-residue-field, cited at proof lines 10986-10988. Produces the open U on which E has the stated form.
- The two defining compositions of the localized class. Same chapter, Lemma labelled lemma-loc-chern-disjoint and Lemma labelled lemma-loc-chern-after-pushforward, cited at proof lines 11004-11008. Give the vanishing against i_{1,*} and the identification against i_{2,*}.

Source reference. Original lemma 0F9H: sources/stacks-chow/pinned-original-source.tex, source lines 9695–9695. Same-chapter pointer into the author environment lemma at source line 9695; invoked at proof line 10998 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original definition 0FB5: sources/stacks-chow/pinned-original-source.tex, source lines 10787–10787. Same-chapter pointer into the author environment definition at source line 10787; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0GUJ: sources/stacks-chow/pinned-original-source.tex, source lines 10678–10678. Same-chapter pointer into the author environment situation at source line 10678; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BCD: sources/stacks-chow/tags.txt, source lines 14276–14276. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 10990 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FB7: sources/stacks-chow/pinned-original-source.tex, source lines 10956–10956. Same-chapter pointer into the author environment lemma at source line 10956; invoked at proof line 10995 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FB6: sources/stacks-chow/pinned-original-source.tex, source lines 10940–10940. Same-chapter pointer into the author environment lemma at source line 10940; invoked at proof line 11004 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FB4: sources/stacks-chow/pinned-original-source.tex, source lines 10918–10918. Same-chapter pointer into the author environment lemma at source line 10918; invoked at proof line 11008 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Lemma 0F9H say E_2 is the restriction of a perfect $E \in D(\mathcal{O}_X)$ whose restriction to X_1 is zero, resp. isomorphic to a finite locally free $\mathcal{O}_{X_1}$-module of rank $< p$ sitting in cohomological degree 0. Then the class $P'_p(E_2)$, resp. $c'_p(E_2)$ of Lemma 0F9H agrees with $P_p(X_2 \to X, E)$, resp. $c_p(X_2 \to X, E)$ of Definition 0FB5 provided E satisfies assumption (3) of Situation 0GUJ.*

Proof. The assumptions on E imply that there is an open $U \subset X$ containing X_1 such that $E|_U$ is zero, resp. isomorphic to a finite locally free $\mathcal{O}_U$-module of rank $< p$. See More on Algebra, Lemma 0BCD. Let $Z \subset X$ be the complement of U in X endowed with the reduced induced closed subscheme structure. Then $P_p(X_2 \to X, E) = (Z \to X_2)_* \circ P_p(Z \to X, E)$, resp. $c_p(X_2 \to X, E) = (Z \to X_2)_* \circ c_p(Z \to X, E)$ by Lemma 0FB7. Now we can prove that $P_p(X_2 \to X, E)$, resp. $c_p(X_2 \to X, E)$ satisfies the characterization of $P'_p(E_2)$, resp. $c'_p(E_2)$ given in Lemma 0F9H. Namely, by the relation $P_p(X_2 \to X, E) = (Z \to X_2)_* \circ P_p(Z \to X, E)$, resp. $c_p(X_2 \to X, E) = (Z \to X_2)_* \circ c_p(Z \to X, E)$ just proven and the fact that $X_1 \cap Z = \emptyset$, the composition $P_p(X_2 \to X, E) \circ i_{1,*}$, resp. $c_p(X_2 \to X, E) \circ i_{1,*}$ is zero by Lemma 0FB6. On the other hand, $P_p(X_2 \to X, E) \circ i_{2,*} = P_p(E_2)$, resp. $c_p(X_2 \to X, E) \circ i_{2,*} = c_p(E_2)$ by Lemma 0FB4. $\square$